LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 4 Carbon and its Compounds

Chapter / Topic: Carbon and its Compounds · Week 2 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each. Do Day 1 on the day you study the chapter, then Day 2 and Day 3 on the next two days.
2. Write your answers in your notebook, then check them with the key the same day. Copy every mistake into your error log.
3. Draw every structure with all its bonds. Check that each carbon has exactly four bonds!

DAY 1: BONDING AND STRUCTURES

Q1. How many electrons are shared between the two nitrogen atoms in a molecule of N_2?

(a) 2 (b) 4
(c) 6 (d) 8

Q2. Which form of carbon is made of football-shaped molecules of 60 carbon atoms?

(a) diamond (b) graphite
(c) charcoal (d) buckminsterfullerene

Q3. The molecular formula of cyclohexane is:

(a) C_6H_6 (b) C_6H_{12}
(c) C_6H_{14} (d) C_6H_{10}

Q4. The general formula of the alkynes is:

(a) C_nH_{2n+2} (b) C_nH_{2n}
(c) C_nH_{2n-2} (d) C_nH_n

Q5. Draw the electron-dot structures of (a) an oxygen molecule, O_2, and (b) carbon dioxide, CO_2.

Q6. Diamond and graphite are both pure carbon. Why is diamond very hard, while graphite is soft and slippery?

Q7. Write the structural formula and the molecular formula of (i) propane, (ii) propene and (iii) propyne.

Q8. Draw the structures of cyclohexane and benzene. How is the bonding different?

DAY 2: FUNCTIONAL GROUPS, NAMING AND ISOMERS

Q9. The IUPAC name of $CH_3–CH_2–CH_2–COOH$ is:

(a) propanoic acid (b) butanal
(c) butanol (d) butanoic acid

Q10. Which pair belongs to the same homologous series?

(a) CH_4 and C_2H_4 (b) C_2H_6 and C_3H_8
(c) C_2H_4 and C_2H_2 (d) CH_3OH and CH_3COOH

Q11. The formula of chloropropane is:

(a) C_3H_7Cl (b) C_3H_8Cl
(c) C_3H_6Cl (d) C_2H_5Cl

Q12. Name the functional group in each: (a) CH_3CHO (b) $CH_3COCH_2CH_3$ (c) CH_3CH_2COOH (d) CH_3Cl.

Q13. Draw the structures of (a) ethanoic acid, (b) bromopentane and (c) butanone.

Q14. Pentane, C_5H_{12}, has three structural isomers. Draw all three.

Q15. Carbon forms neither C^{4+} nor C^{4-} ions. Explain why, and say how carbon achieves a stable configuration instead.

DAY 3: REACTIONS, ETHANOL AND ETHANOIC ACID

Q16. **Assertion (A):** Oils containing unsaturated fatty acids should be chosen for cooking.
Reason (R): Animal fats generally contain saturated fatty acids, which are said to be harmful for health.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q17. Denatured alcohol is ethanol made unfit for drinking by adding:

(a) methanol and a blue dye
(b) only water
(c) ethanoic acid
(d) glucose

Q18. Which of these reacts with chlorine in sunlight by substitution?

(a) C_2H_4
(b) C_2H_2
(c) CH_4
(d) CO_2

Q19. A teacher gave Aryan two unlabelled bottles, one of ethanol and one of ethanoic acid. He added a pinch of sodium hydrogencarbonate to a little of each. One bottle gave brisk effervescence; the other showed no change.

(i) Which liquid gave the effervescence? Write the balanced equation.

(ii) How can he test the gas given off?

(iii) Suggest one more way to tell the two liquids apart.

Q20. (a) Write the balanced equation for the complete combustion of ethanol.
(b) What is an oxidising agent? Name two substances that can oxidise ethanol to ethanoic acid.
(c) What is vinegar? Why is ethanoic acid called a weak acid?

Q21. Compound A has the formula C_2H_6O and is found in alcoholic drinks. When A is heated with excess concentrated sulphuric acid at 443 K, it gives a gas B. B adds hydrogen in the presence of a nickel catalyst to give C, C_2H_6. Identify A, B and C and write the two balanced equations.

~ Week 2 done after this one. Big chapters, great work! ~